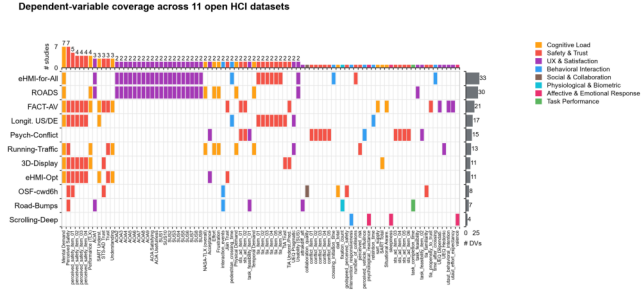
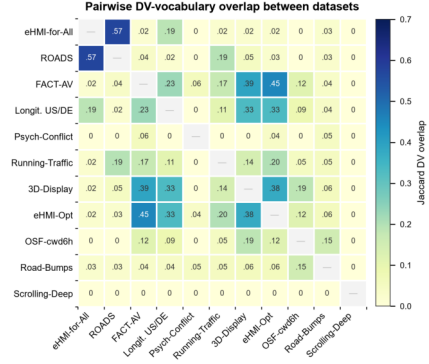
OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

(a) DV coverage across 11 datasets

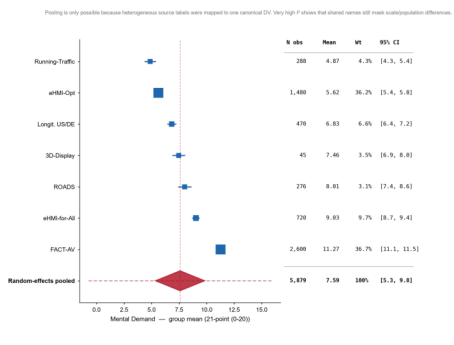


After schema-driven standardization, mean pairwise DV overlap is only Jaccard = 0.093; 12 DVs are shared by >2 studies.

(b) Pairwise overlap (Jaccard)

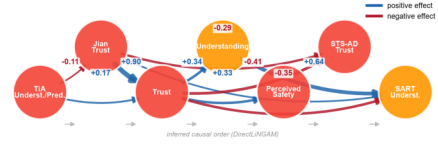


(c) Meta-analysis: Mental Demand



Random-effects (DerSimonian-Laird), k = 7 studies. Heterogeneity: I^2 = 99.7%, τ^2 = 9.12, Q(6) = 2183, p < 0.001
Hartung-Knapp-Sidik-Jonkman 95% CI (t, 6 df): [5.6, 9.6]

(d) Cross-study causal structure



DirectLiNGAM on within-study z-scores (complete-case on the largest co-observed DV set; n = 2,600 obs; all rows from one study (FACT-AV), i.e. a within-study DAG). Edge weights are structural coefficients.